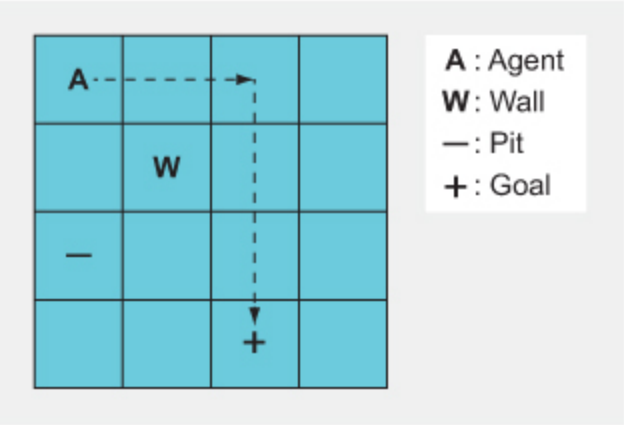
Gridworld board to the goal, following the most efficient route every time. But before we get too far into that, let's review the key terms and concepts from the previous chapter, which we will continue to use here.

Figure 3.1. This is a simple Gridworld game setup. The agent (A) must navigate along the shortest path to the goal tile (+) and avoid falling into the pit (–).



The *state* is the information that our agent receives and uses to make a decision about what action to take. It could be the raw pixels of a video game, sensor data from an autonomous vehicle, or, in the case of Gridworld, a tensor representing the positions of all the objects on the grid.

The *policy*, denoted π, is the strategy our agent follows when provided a state. For example, a policy in Blackjack might be to look at our hand (the state) and hit or stay randomly. Although this would be a terrible policy, the important point to stress is that the policy confers which actions we take. A better policy would be to always hit until we have 19.

The *reward* is the feedback our agent gets after taking an action, leading us to a new state. For a game of chess, we could reward our agent +1 when it performs an action that leads to a checkmate of the other player and –1 for an action that leads our agent to be checkmated. Every other state could be rewarded 0, since we do not know if the agent is winning or not.

Our agent makes a series of actions based upon its policy π, and repeats this process until the episode ends, thereby we get a succession of states, actions and the resulting rewards.